

Analysis of Power Losses in Power MOSFET Based Stacked Polyphase Bridges Converters

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Abstract—The ability to accurately predict power semiconductor losses is essential for converter design and can provide a reference for thermal management. Based on a stacked polyphase bridges converter, a simplified but accurate loss model is used in this paper to calculate the power losses of power MOSFETs and experimentally verified. A special focus of the analysis is the ringing loss calculation since it is inevitable in the switching process of power MOSFETs. In addition, this paper presents a comparison of power losses between Si MOSFET and GaN FET based converters by using the analytical model.

Index Terms—Power losses, stacked polyphase bridges converter, power MOSFET, GaN FET.

I. INSTRUCTION

An accurate power loss model can enable converter designers to make fast calculations, compare between different semiconductors, and size the heatsink properly. As the switching frequency is increased to improve the converter performance, the switching losses of power MOSFETs becomes a dominant factor in the total power losses. Traditional loss models treat the switching waveforms as piecewise linear [1], providing a simple and fast calculation based on datasheet information. However, it does not account for the parasitic inductances and nonlinear characteristics of the parasitic capacitors which significantly affect the switching process in practice. Many detailed analytical models that considering such parasitics have been proposed [2]–[4]. Most of them are based on the buck converter and only provide fully analytical expressions for the main transition period, while the ringing period caused by the resonance between the parasitics is always observed and could result in significant loss.

The stacked polyphase bridges (SPB) converter (see Fig. 1) consists of several series connected two-level three-phase converters, which allows the utilization of low-voltage, high-switching frequency semiconductors [5]. This paper presents an analytical power loss calculation for a power-MOSFETs based SPB converter that includes all the major parasitics and is experimentally verified in a scaled-down prototype. The analytical expressions for both the main transition period and the ringing period are derived by solving the equivalent circuits during each switching process.

Wide-band-gap devices such as silicon carbide (SiC) and gallium nitride (GaN) are gaining more and more interest [6].

The maximum voltages of most GaN devices available in the market are less than 200 V, which is too low for conventional drive systems. Whereas the submodules of the SPB converter are connected in series to reduce the submodule dc voltage, offering an opportunity to employ low-voltage GaN transistors. Based on the analytical loss model, this paper also presents a power loss comparison between Si-MOSFET-based SPB converter and GaN-FET-based SPB converter.

The paper is outlined as follows. In Sections II and III, The loss-thermal model and the switching process is illustrated analytically. In Section IV, experimental results are provided to verify the accuracy of the loss model. Moreover, the loss comparison between different low-voltage, high-switching frequency semiconductors is presented in Section V. Finally, a summary is given in Section VI.

II. LOSS-THERMAL MODEL

Some of the semiconductor parameters are temperature dependent, such as on-state resistance $R_{DS(on)}$, therefore a thermal model is adopted in this paper to interact with loss model. Fig. 2 shows the loss-thermal model diagram. As seen, the power MOSFET model is fed with an instantaneous junction temperature from the thermal model. Every temperature dependant parameters of the semiconductor can be calculated from this temperature. Then, the loss model uses these parameters to estimate the power dissipation. The dissipated power in the MOSFET will be delivered again to the thermal model for a new temperature calculation. These

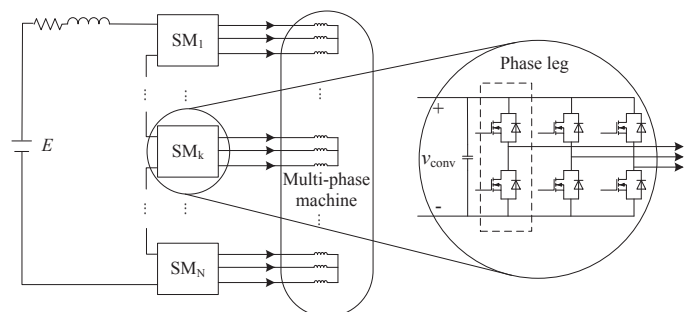


Fig. 1. Structure of the SPB converter topology.

steps iterate until the temperature tolerance is reached.

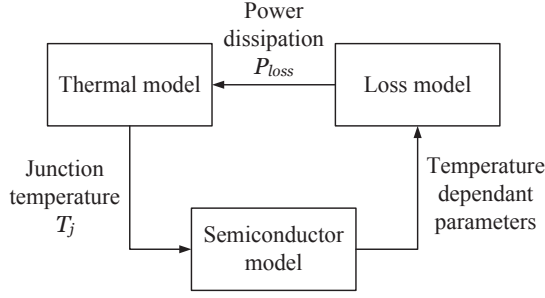


Fig. 2. Loss-thermal model diagram.

In the thermal model, an equivalent thermal network can be utilized (see Fig. 3). The semiconductor device can be modeled as a R-C ladder network, in which the power dissipation is represented as current and the temperature is represented as voltage [7]. In Fig. 3, the symbols $R_{th,jc}$, $R_{th,ch}$ and $R_{th,ha}$ stand for junction-to-case, case-to-heatsink, heatsink-to-ambient thermal resistances, respectively. C_{th} stands for thermal capacitance; can be neglected in static thermal network.

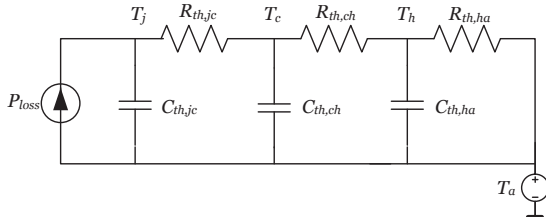


Fig. 3. Equivalent thermal network.

III. CONVERTER LOSS ANALYSIS

For power MOSFETs, the calculation of conduction losses is quite straightforward by calculating $I_{drms}^2 R_{DS(on)}$. Therefore, only the switching loss calculation will be presented in the following analysis [8]. Fig. 5(a) shows a simplified equivalent circuit of one phase leg from Fig. 1, including the most relevant parasitics present in actual power converters, such as parasitic inductances in series with drain and source terminals of the MOSFET, and inherent parasitic capacitances C_{gs} , C_{gd} , C_{ds} . These parasitics significantly affect the switching process in practice, for instance, the ringing in Fig. 4.

As mentioned in Section II, the increase of $R_{DS(on)}$ with junction temperature T_j is taken into account in the thermal model, as is the nonlinearity of the parasitic capacitances with v_{ds} in the loss model; these data can be extracted from the datasheet.

A. Turn-on Transition

Fig. 5(b)-(e) shows the equivalent circuits at different periods of the turn-on process.

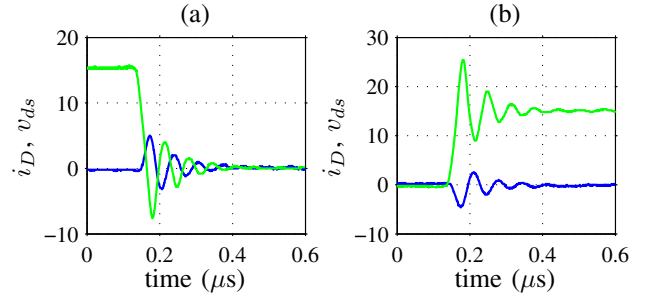


Fig. 4. Typical switching waveforms of power MOSFETs: (a) turn-on. Green: v_{ds} ; blue: i_D . (b) turn-off. Green: v_{ds} ; blue: i_D .

1) *Period I*: For simplicity, a freewheeling diode is used instead of the bottom switch during the commutation period. Given the commutation time is sufficiently short, the load current can be treated as a dc current source I_{load} . Therefore, Fig. 5(a) can be further simplified as Fig. 5(b), in which $L_s = L_{dl} + L_{sl}$, $L_d = L_{du} + L_{su}$.

At t_0 , the gate voltage V_{dr} is added to the top switch, charging the input capacitances C_{gd} and C_{gs} . This period ends when the gate voltage v_{gs} has risen to the threshold voltage V_{th} at t_1 . At this substage, the drain current i_d of the top switch is still zero; there is no power loss during this period.

2) *Period II*: At t_1 , the drain current i_d starts to take over the load current. Based on Fig. 5(c), we can obtain the following equations on Laplace form

$$v_{ds}(s) = V_{conv} - sL_d i_D(s) - sL_s(i_D(s) - I_{load}) \quad (1)$$

$$v_{gs}(s) = v_{ds}(s) + v_{gd}(s) \quad (2)$$

$$i_G(s) = (V_{dr} - v_{gs}(s))/R_g = sC_{gd}v_{gd}(s) + sC_{gs}v_{gs}(s) \quad (3)$$

$$i_D(s) = i_d(s) - sC_{ds}v_{ds}(s) - sC_{gd}v_{gd}(s) \quad (4)$$

$$i_d(s) = g_{fs}(v_{gs}(s) - V_{th}) \quad (5)$$

where g_{fs} is the transconductance of a MOSFET and assumed to be constant in this paper. Then the gate-source voltage v_{gs}

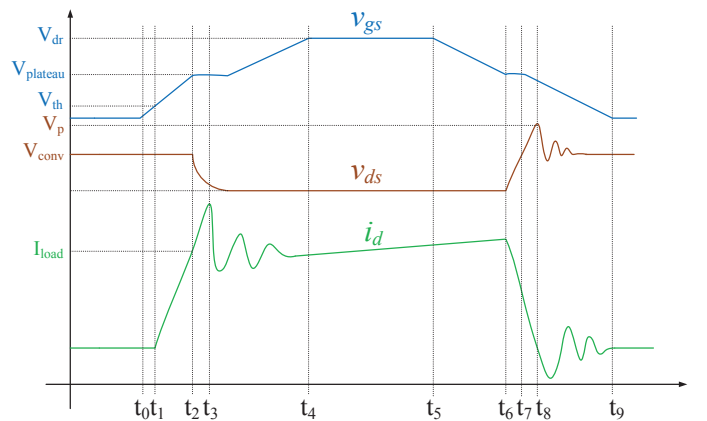


Fig. 6. Key waveforms of the switching transitions.

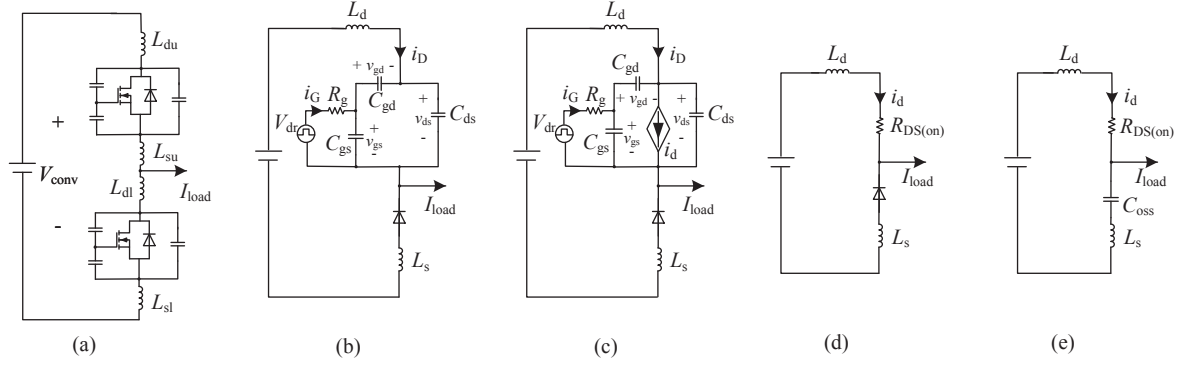


Fig. 5. Simplified equivalent circuits during turn-on period of the top switch. (a) equivalent circuit of a phase leg; (b) period I: delay period; (c) period II: turn-on transition; (d) period III: reverse recovery; (e) period III: ringing.

is derived as

$$v_{gs}(s) = \frac{V_{dr}}{q_1 s^2 + q_2 s + 1}$$

where

$$q_1 = L_{eq} C_{eq2} + R_g g_{fs} C_{gd} L_{eq} \quad (7a)$$

$$q_2 = R_g C_{eq1} \quad (7b)$$

$$L_{eq} = L_s + L_d \quad (7c)$$

$$C_{eq1} = C_{gd} + C_{gs} \quad (7d)$$

$$C_{eq2} = C_{gd} - C_{ds}. \quad (7e)$$

The drain-source voltage and drain current can be obtained from (2) and (5)

$$v_{ds}(s) = \frac{-V_{dr} q_1 s}{R_g C_{gd} (q_1 s^2 + q_2 s + 1)} \quad (8)$$

$$i_d(s) = g_{fs} \left(\frac{V_{dr}}{q_1 s^2 + q_2 s + 1} - V_{th} \right). \quad (9)$$

Transferring to time domain gives the solution of $v_{ds}(t)$ and $i_d(t)$. The energy dissipation during this period is

$$E_{turn_on} = \int_{t_1}^{t_2} v_{ds}(t) i_d(t) dt. \quad (10)$$

The drain current reaches the load current I_{load} at t_2 .

3) *Period III*: At t_2 , the top switch takes over the load current I_{load} and the bottom diode starts to recover, but still can not block the voltage. i_d reaches its peak value at t_3 when the diode starts to block voltage. Beyond t_3 , the ringing continues and assumed to be completely damped at t_4 . The equivalent circuits for $[t_2, t_3]$ and $[t_3, t_4]$ are shown in Figs. 5(c) and (d), respectively. The drain current i_d during ringing period is expressed using Laplace form

$$i_d(s) = \frac{C_{oss} v_{C_{oss}}(0_-) - s L_{eq} C_{oss} i_{L_{eq}}(0_-)}{L_{eq} C_{oss} s^2 + R_{DS(on)} C_{oss} s + 1} \quad (11)$$

where $v_{C_{oss}}(0_-)$ is the forward voltage of the body diode and $i_{L_{eq}}(0_-) = I_{load}$.

By considering the energy stored at the beginning and the

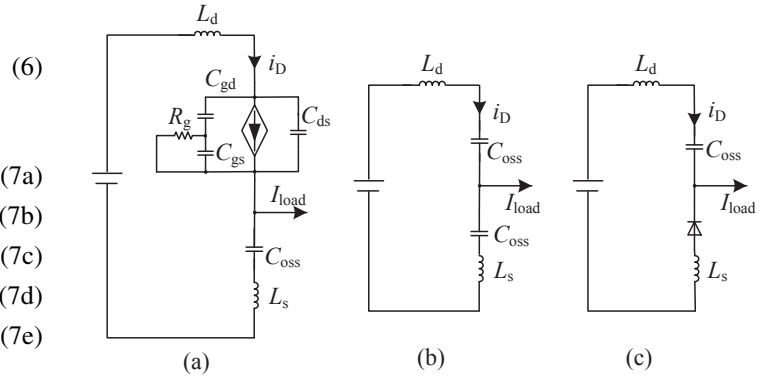


Fig. 7. Simplified equivalent circuits during turn-off period of the top switch. (a) period II: turn-off transition; (b) period II: v_{ds} continue rising; (c) period III: ringing.

end of $[t_3, t_4]$

$$E_{store_t3} = \int_{t_2}^{t_3} V_{conv} (i_d(t) - I_{load}) dt \quad (12)$$

$$E_{store_t4} = \frac{1}{2} C_{oss} V_{conv}^2 \quad (13)$$

the dissipated ringing energy is

$$E_{ring_on} = E_{store_t3} - E_{store_t4}. \quad (14)$$

B. Turn-off Transition

Fig. 7(a)-(c) shows the equivalent circuits at different periods of the turn-off transition.

1) *Period I*: At t_5 , the gate circuit starts to discharge the capacitances C_{gs} and C_{gd} , causing the gate-source voltage v_{gs} starts to fall. But v_{ds} and i_d do not change in this period. This period ends when v_{gs} decreases to the plateau voltage $V_{th} + I_{load}/g_{fs}$ at t_6 . Power losses can be neglected during $[t_5, t_6]$ as this period is very short.

2) *Period II*: At t_6 , the drain-source voltage v_{ds} starts to rise while the gate voltage is held at the plateau level. The excess current $I_{load} - i_d$ charges the capacitances. i_d does not change until v_{ds} reaches the input voltage V_{conv} at t_7 . After

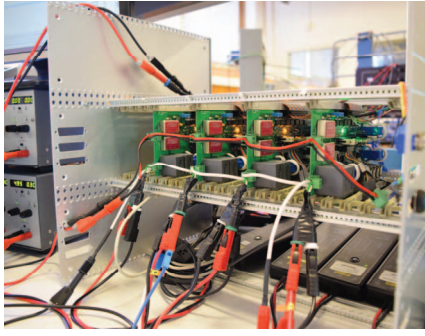


Fig. 8. Four-submodule SPB converter.

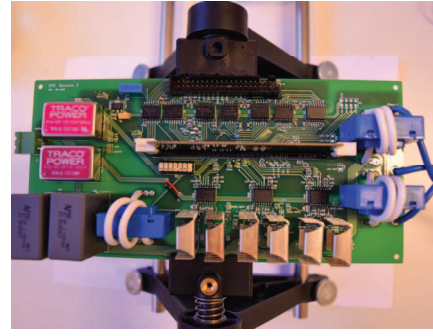


Fig. 9. Single submodule.

that, v_{ds} continues rising due to the existence of the parasitics. Similar to Section III-A2, by solving circuit equations based on Fig. 7(a), the gate-source voltage v_{gs} on Laplace form is

$$v_{gs}(s) = \frac{V_{th} + I_{load}/g_{fs}}{q_1 s^2 + q_2 s + 1} \quad (15)$$

where q_1 and q_2 are the same as in Section III-A2.

This period ends at t_8 when the drain current reaches zero. At this moment, the drain-source voltage reaches its peak voltage V_p . Therefore, during $[t_6, t_8]$ the dissipated turn-off energy is

$$E_{turn_off} = \int_{t_6}^{t_8} v_{ds}(t) i_d(t) dt. \quad (16)$$

3) *Period III*: A voltage ringing period continues after t_8 until v_{ds} reaches its steady-state value at t_9 . Based on the equivalent circuits shown in Fig. 7, the drain current i_d can be derived and is the same as (11) with $v_{C_{oss}}(0_-) = V_{conv}$ and $i_{Leq}(0_-) = 0$. Similar to Section III-A3,

$$E_{store_t_8} = \frac{1}{2} C_{oss} V_p^2 + \frac{1}{2} L_s I_{load}^2 \quad (17)$$

$$E_{store_t_9} = \frac{1}{2} C_{oss} V_{conv}^2 + \frac{1}{2} L_s I_{load}^2. \quad (18)$$

losses dissipated in this period is obtained as

$$E_{ring_off} = E_{store_t_8} - E_{store_t_9} - V_{conv} \int_{t_8}^{t_9} i_d(t) dt \quad (19)$$

$$= \frac{1}{2} C_{oss} (V_p^2 - V_{conv}^2) - V_{conv} \int_{t_8}^{t_9} i_d(t) dt. \quad (20)$$

IV. MODEL VERIFICATION

In order to experimentally test the analytical model, laboratory measurements were conducted on one submodule of an SPB converter prototype (see Figs. 8 and 9). In this prototype, the selected MOSFET is Infineon IPP075N15N3G (150 V/100 A). The main parameters are listed in Table I.

It can be seen from Fig. 10, in which the dc component of the switching current is not included, the ringing waveforms during turn-on and turn-off transitions can be well predicted according to (11). This indicates the accuracy of the presented model in the ringing period.

The comparison of measured loss and calculated power loss

as a function of output current and switching frequency is shown in Fig. 11. The switching frequency varies from 10 kHz to 40 kHz. The modulation index is fixed at 0.9 for all cases. As seen, the calculated results match the experimental results well. It shows good accuracy and can be used to predict the power dissipation in the converter design process.

V. COMPARATIVE EVALUATION

As mentioned before, low-voltage GaN FETs can be good alternative for the SPB converter, therefore, a comparative evaluation between the GaN-FET-based SPB converter and the Si-MOSFET-based SPB converter is presented in this section. Considering a four-submodule SPB converter in a typical automotive drive system (with 400 V dc-link voltage), similar rated eGaN FET EPC2033 (150 V/31 A) from Efficient Power Conversion Corporation (EPC) and Si MOSFET

TABLE I PARAMETERS OF THE SETUP UNDER TEST

Parameter	Value	Unit	Symbol
Dc power supply	300	V	E
Number of submodules	4	-	N
Submodule dc capacitance	94	μF	C
Load resistance	2	Ω	R_{ac}
Load inductance	4.67	mH	L_{ac}

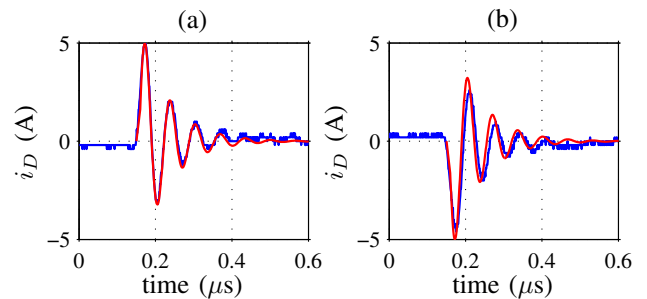


Fig. 10. Analytical and experimental ringing waveforms: (a) turn-on; (b) turn-off. Red: analytical; blue: measured.

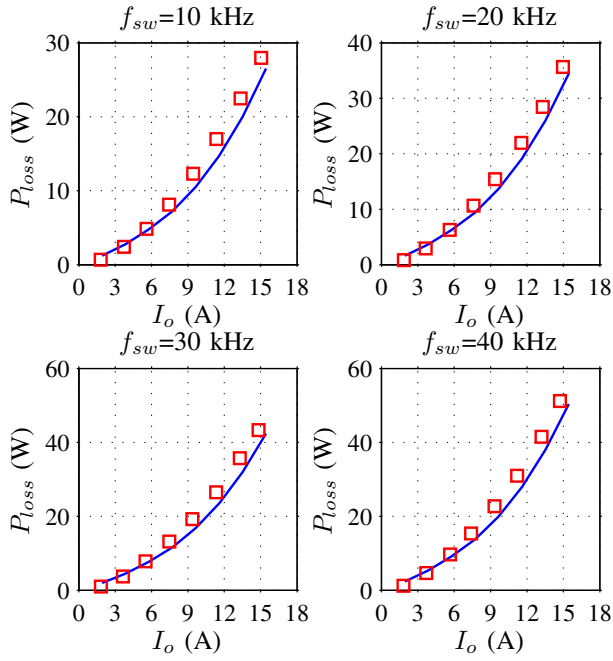


Fig. 11. Loss model verification with experimental results. Red: experiment; blue: calculation.

IPA105N15N3G (150 V/37 A) from Infineon were selected for the comparison.

A. Comparison of eGaN FET and Si MOSFET

Table II shows main parameters of EPC2033 and IPA105N15N3G. As seen, eGaN FET has similar $R_{DS(on),max}$, C_{oss} , and C_{rss} with the Si MOSFET, but a much lower input capacitance. Moreover, the reverse recovery charge of EPC2033 is zero. For eGaN FETs, once the threshold voltage V_{th} is reached, there will be sufficient electrons under the gate to form a conductive channel. Therefore, there is no reverse recovery losses in eGaN FETs [9], which indicates that the switching loss could be lower than Si MOSFETs with similar ratings.

TABLE II MAIN PARAMETERS OF eGaN FET AND Si MOSFET

Parameter	eGaN FET	Si MOSFET
Product	EPC2033	IPA105N15N3G
On-resistance $R_{DS(on),max}$ [mΩ]	7	10.5
Input capacitance C_{iss} [pF]	1140	3233
Output capacitance C_{oss} [pF]	580	378
Reverse transfer capacitance C_{rss} [pF]	8	7
Reverse recovery charge Q_{rr} [nC]	0	380

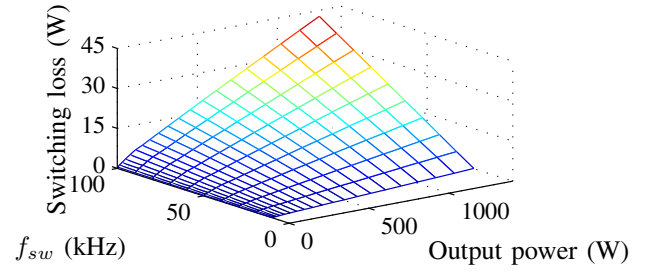


Fig. 12. Switching losses as a function of switching frequency and output power for Si-MOSFET-based converter.

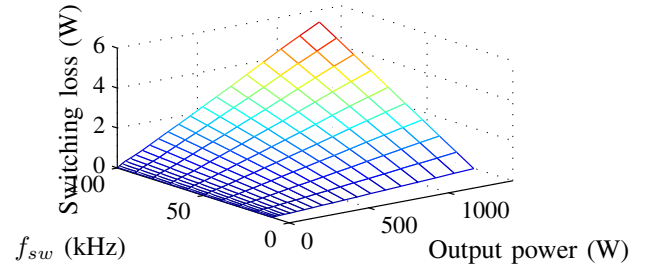


Fig. 13. Switching losses as a function of switching frequency and output power for eGaN-FET-based converter.

B. Loss Comparison of eGaN-FET and Si-MOSFET based SPB Converter

According to [9], eGaN FETs from EPC behave very similarly to the Si MOSFETs. The presented analytical model thus can be used to estimate the power losses in EPC's eGaN FETs.

Figs. 12 and 13 show a switching loss comparison of the selected semiconductors as a function of switching frequency and output power at a fixed modulation index of 0.9. It can be seen that the calculated results are in accordance with the expectation from above discussion. The switching loss of eGaN-FET-based SPB converter is about 15% of Si-MOSFET-

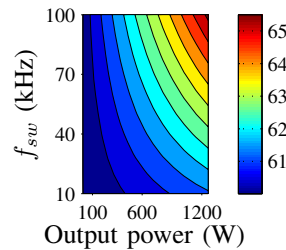


Fig. 14. Junction temperature of Si MOSFET.

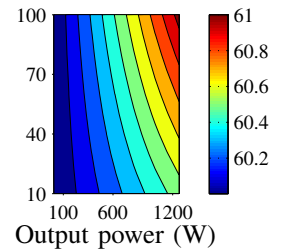


Fig. 15. Junction temperature of eGaN FET.

based SPB converter.

The thermal comparison is presented in Figs. 14 and 15. Ambient temperature is set to 60°C. It should be pointed out that the calculations are made under the assumption of liquid cooling with a low thermal resistance from the case to the cooling water. The junction temperature rise of eGaN FET is about 1/6 of Si MOSFET, which implies the reduction of radiating equipment and improvement of power density.

VI. CONCLUSION

In this paper, power losses of the SPB converter was analyzed and experimentally investigated. The analytical model provides a well estimation of the switching loss. Based on this model, a comparative evaluation between the GaN-FET-based SPB converter and the Si-MOSFET-based SPB converter was presented. GaN-based converter operated up to 100 kHz switching frequency with 0.5% switching losses at 1.4 kW output power, while Si-MOSFET-based converter dissipated 3.2% switching losses at the same condition. Furthermore, the radiating equipment of the GaN-based converter can be reduced due to smaller temperature rise, which indicates a more compact hardware system.

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